

Question -1

1. Clone <https://github.com/zembutsu/docker-sample-nginx.git> on your local system.
 - Initialize one repo on your local system named Cloudethix-nginx-yourname and then copy all the content from cloned repo to your local repo.
 - Update the index.html and add the name CloudEthiX along with Hostname. Commit the change to your local repository. Then create and push the code from the local repository to your remote repository named Cloudethix-nginx-yourname under master branch.
 - Once code is pushed, create a Docker hub public repository named Cloudethix-nginx-yourname.
 - Then build the docker image out of the dockerfile. Once an image is built, tag the image and push that image to the docker hub repo you created.
 - Create a deployment.yaml file with 2 replications in cloudethix_yourname Namespace and use the container image that you have pushed to your docker hub repo. Once a file is created, run the deployment in k8s cluster.
 - Please prepare a document along with the snap and share it across

System Output

The following output shows the commands executed and confirms that the required Git and Kubernetes tasks were completed.

```
payal_kharat@LAPTOP-# git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
payal_kharat@LAPTOP-# mkdir cloudethix-nginx-payal
payal_kharat@LAPTOP-# cd cloudethix-nginx-payal/
-----
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/assignment-1/cloudethix-nginx-payal/.git/
```

```
payal_kharat@LAPTOP-# cp -rf docker-sample-nginx/* cloudethix-nginx-payal/
payal_kharat@LAPTOP-# cd cloudethix-nginx-payal/
payal_kharat@LAPTOP-# ls
Dockerfile  LICENSE  README.md  default.conf  index.html
```

```
payal_kharat@LAPTOP-# git status
```

On branch master

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

Dockerfile

LICENSE

README.md

default.conf

index.html

nothing added to commit but untracked files present (use "git add" to track)

```
payal_kharat@LAPTOP-# git add .
```

```
payal_kharat@LAPTOP-# git commit -m "updated index.html file"
```

[master (root-commit) 9c3f2e4] updated index.html file

5 files changed, 48 insertions(+)

create mode 100644 Dockerfile

create mode 100644 LICENSE

create mode 100644 README.md

create mode 100644 default.conf

create mode 100644 index.html

```
payal_kharat@LAPTOP-# git pull origin master --allow-unrelated-histories --no-rebase
```

From https://github.com/CloudEthix987/Cloudethix-Nginx-Payal

* branch master -> FETCH_HEAD

Auto-merging index.html

CONFLICT (add/add): Merge conflict in index.html

Automatic merge failed; fix conflicts and then commit the result.

```
payal_kharat@LAPTOP-# git status
```

On branch master

You have unmerged paths.

(fix conflicts and run "git commit")

(use "git merge --abort" to abort the merge)

Unmerged paths:

(use "git add <file>..." to mark resolution)

both added: index.html

no changes added to commit (use "git add" and/or "git commit -a")

```
payal_kharat@LAPTOP-# git add .
```

```
payal_kharat@LAPTOP-# git status
```

On branch master

All conflicts fixed but you are still merging.

(use "git commit" to conclude merge)

Changes to be committed:

modified: index.html

```
payal_kharat@LAPTOP-# git commit -m "Index.html is updated 2nd time"
```

[master 3558b9c] Index.html is updated 2nd time

```
payal_kharat@LAPTOP-# git push -u origin master
```

Username for 'https://github.com': CloudEthix987

Password for 'https://CloudEthix987@github.com':

Enumerating objects: 8, done.

Counting objects: 100% (8/8), done.

Delta compression using up to 12 threads

Compressing objects: 100% (6/6), done.

Writing objects: 100% (6/6), 710 bytes | 44.00 KiB/s, done.

```
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/CloudEthix987/Cloudethix-Nginx-Payal.git
    9e6a3cb..3558b9c  master -> master
branch 'master' set up to track 'origin/master'.

-----

payal_kharat@LAPTOP-# vim deployment.yaml
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/cloudethix-nginx created
payal_kharat@LAPTOP-# kubectl get pods -n cloudethix-payal
NAME                                READY   STATUS    RESTARTS   AGE
cloudethix-nginx-84b5b9996f-r5dsm   1/1     Running   0           38s
cloudethix-nginx-84b5b9996f-s2v69   1/1     Running   0           38s
payal_kharat@LAPTOP-# kubectl get deployment -n cloudethix-payal
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
cloudethix-nginx    2/2     2             2           54s
```